Kenneth Chen Ko Han

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Education

National University of Singapore Aug 2026 — May 2029
B.Comp. in Computer Science (NUS College), Minor in Mathematics *Singapore*
- **NUS Merit Scholarship** – awarded to high-calibre freshmen on academic results, leadership qualities and co-curricular record.

Singapore Polytechnic Apr 2021 — Apr 2024
Diploma in Applied AI and Analytics, CGPA: 4.0/4.0 *Singapore*
- **Lee Kuan Yew Award for Mathematics & Science** – top tech graduate, **IMDA Gold Medal** – top all-around achiever, **SP Scholarship**.
- Coursework: Deep Learning, Machine Learning, Reinforcement Learning, Full-Stack Development, DevOps, Data Analytics, Data Engineering and Visualisation.

Work Experience

Team Lead Oct 2026 — Present
NUS College Club *Singapore*
- Led a team of 3 engineers and 1 PM to ship the NUS College end-of-semester dinner ticketing system (ticket.nusc.club) for 900+ attendees.
- Built QR-code check-in with atomic updates and encryption of student data.
- Ensured compliance with NUS data privacy policy, making the case to NUS College administration on the system's security and reliability.

Co-founder and AI Lead Nov 2023 — Aug 2026
Resumify *Singapore*
- Non-profit built with YellowRibbon.gov.sg, HTX and the Singapore Prison Service (resumify.org); AI-powered resume services delivered to 600+ users.
- Awarded SGD 24,000 in funding by the National Youth Council; signed a 12-month MOU with YellowRibbon; featured in The Business Times and Lianhe Zaobao.
- Collaborated with SPD (Society for the Physically Disabled); hosted a sharing session to train career facilitators on using the platform.
- Stack: Next.js, LLM agents, OpenAI API, ElevenLabs, Vercel, Render, Supabase, SQL. Developed with OpenCode and Semble Search.

AI Engineer (Volunteer) Oct 2023 — Jun 2026
Republic of Singapore Air Force (RAiD) *Singapore*
- Built the full-stack system that analyses Air Traffic Control situations: an NVIDIA NeMo speech recognition and speaker diarisation layer (fine-tuned parakeet-tdt, canary-1b and NGPU-LM) feeding a Qwen3 reasoning layer served with vLLM.
- Deployed it to edge devices for real-time transcription in the control room; presented to the Chief of Air Force.
- Built an LLM-powered internal search tool at a RAiD hackathon hosted at Microsoft's office.

AI Intern Apr 2024 — Jun 2024
Bifrost AI *Singapore*
- Series A startup building synthetic data solutions for AI companies, backed by Sequoia Capital.
- Trained YOLOv6-lite for military tank detection with ST Engineering's Unmanned Vehicles Team; tuning for incremental learning lifted accuracy 16%+.
- Fine-tuned a StabilityAI VAE to improve text and high-frequency reconstruction for synthetic-to-real domain adaptation.
- Prototyped a zero-shot method with JinaCLIP to visualise missing data for NASA, BigBear AI and DSTA.

AI Engineer Intern Apr 2023 — Mar 2024
GovTech Singapore *Singapore*
- Video Analytics team, Data Science and AI Division. Lead AI engineer on a computer vision pipeline for MSO's OneService App on AWS (EKS, SageMaker), classifying thousands of daily municipal reports via zero-shot ensembles.
- Built real-time animal welfare detection for NParks with OWLv2, tuned by Bayesian optimisation (Hyperopt); the pitch secured executive-board funding.
- Red-teamed GovTech's AI Image Detector with custom-trained LoRAs simulating locally contextualised deepfakes.
- Prototyped a novel feature-explainability method for object detection using SegGPT and SAM.

Projects

Air Traffic Control Automatic Speech Recognition (ATC-ASR) (huggingface.co/qenneth) 2025
- Fine-tuned parakeet-tdt-0.6b-v3 with NVIDIA NeMo to a state-of-the-art 5.99% Word Error Rate on the Jacktol ATC-ASR test split, trained in under an hour on a single H200; 200+ Hugging Face downloads a month at peak, now averaging 70.

Publications

Crowdsource, Crawl, or Generate? Creating SEA-VL, a Multicultural Vision-Language Dataset for Southeast Asia ACL 2025
- Co-author on a 100+ author SEACrowd collaboration; annotated Myanmar-specific imagery and contributed to the writing.

Awards and Competitions

TCS CodeVita S10 – 8th of 100,000+ students globally; 17th of 30 at the Grand Finale, Hyderabad 2023
USA Computing Olympiad – Gold Division 2025
DSTA BrainHack CODE_EXP – finalist; built a "pocket data scientist" advising hawkers on sales and demand trends 2024

Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, HTML, CSS, Bash
Machine Learning: PyTorch, Hugging Face, NVIDIA NeMo, vLLM, scikit-learn, NumPy, Pandas, OpenCV, LoRA, Hyperopt, Weights & Biases
Web and Data: Next.js, React, FastAPI, Node.js, Tailwind CSS, MySQL, MongoDB, Supabase, OpenAI API
Cloud and Tooling: AWS (EKS, SageMaker), Docker, Vercel, Render, Git, LaTeX, Typst